A MID-TERM PROGRESS REPORT ON

AI-Enabled Hydrological Information System for the Mahanadi Basin

Prithvi Flood-IN: Pre- and During-Event Inundation Mapping with Prithvi-EO-2.0 over the Indian Subcontinent

Submitted in partial fulfilment of the requirements of the course

CE 491: Integrated Design Project

for the degree of

Bachelor of Technology in Civil Engineering

by

Nikhil Upadhyay (Team Leader)	23B0764
Prakhar Gupta	23B0771
Prabhat Kumar	23B0660
Vineet Jangir	23B0762
Himanshu Shete	23B0770

Under the guidance of

Prof. Basudev Biswal
Faculty Supervisor
Department of Civil Engineering
IIT Bombay

Mr. Vivek Gupta
Industry Mentor
Chief Technology Officer
ClimMaTech Pvt. Ltd., IIT Mandi

Department of Civil Engineering
Indian Institute of Technology Bombay
Powai, Mumbai – 400 076

October 2026

Certificate

This is to certify that the mid-term progress report entitled **"AI-Enabled Hydrological Information System for the Mahanadi Basin — Prithvi Flood-IN: Pre- and During-Event Inundation Mapping with Prithvi-EO-2.0 over the Indian Subcontinent"**, submitted by Nikhil Upadhyay (23B0764), Prakhar Gupta (23B0771), Prabhat Kumar (23B0660), Vineet Jangir (23B0762) and Himanshu Shete (23B0770), is a record of work carried out by them under our supervision as part of the course CE 491: Integrated Design Project in the Department of Civil Engineering, Indian Institute of Technology Bombay, during the Autumn Semester 2026–27.

Prof. Basudev Biswal	**Mr. Vivek Gupta**
Faculty Supervisor	Industry Mentor
Department of Civil Engineering	CTO, ClimMaTech Pvt. Ltd.
IIT Bombay	IIT Mandi, Himachal Pradesh

Date: _____ Place: Mumbai

Declaration

We declare that this written submission represents our ideas in our own words and, where others' ideas or words have been included, we have adequately cited and referenced the original sources. We declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented, fabricated or falsified any idea, data, fact or source in our submission. Work carried out by the industry mentors and reproduced in this report for comparison (notably the benchmark notebook discussed in Chapter 5) is explicitly attributed. We understand that any violation of the above will be cause for disciplinary action by the Institute.

Name	**Roll No.**	**Signature**
Nikhil Upadhyay	23B0764	_____
Prakhar Gupta	23B0771	_____
Prabhat Kumar	23B0660	_____
Vineet Jangir	23B0762	_____
Himanshu Shete	23B0770	_____

Date: _____

Acknowledgements

We express our sincere gratitude to our faculty supervisor, **Prof. Basudev Biswal**, Department of Civil Engineering, IIT Bombay, for giving us the opportunity to work on a problem of real societal relevance, and for his guidance on the hydrological framing of the project. We thank our industry mentor, **Mr. Vivek Gupta**, Chief Technology Officer, ClimMaTech Pvt. Ltd., and **Mr. Siddik Barbhuiya** of the ClimMaTech team, for providing the physics-informed neural network code base, the TerraTorch and TerraMind benchmark material, and continuous technical feedback during weekly reviews. We are grateful to **Mr. Vijay [TA full name]**, teaching assistant, for organising the review meetings and sharing the reading material on inundation, flood routing and runoff generation.

We acknowledge the Head of the Department, **Prof. Tom V. Mathew**, and the Departmental Undergraduate Committee (DUGC) for facilitating industry-linked design projects. We also acknowledge the open data and open-source software that made this work possible: the European Space Agency's Copernicus Sentinel-2 programme, the National Database for Emergency Management (NDEM) of ISRO/NRSC, IBM and NASA for releasing the Prithvi-EO-2.0 foundation model, and the developers of TerraTorch, PyTorch and PyTorch Lightning. Finally, we thank the staff at Pawna Dam for the field visit.

Abstract

Floods are the most frequent natural hazard in India, and rapid, reliable maps of inundated area are essential for during-event response. Official flood maps are still produced largely through expert interpretation, and physics-based hydrodynamic models are limited by the cost and uncertainty of river-geometry and elevation data. This Integrated Design Project, carried out with ClimMaTech Pvt. Ltd. as part of an AI-enabled hydrological information system for the Mahanadi basin, adapts a geospatial foundation model to Indian flood conditions and links it to physically based flood-routing analysis.

In the first half of the project we (i) curated an India-specific flood segmentation dataset by pairing Sentinel-2 six-band surface-reflectance imagery with NDEM flood-extent maps; screening 1,022 NDEM layers produced 180 matched scene–label pairs covering 100 flood events (2021–2026) and 3,584 chips of 512×512 pixels, with clouds, shadows and permanent water bodies assigned an *ignore* label and an event-wise train/validation/test split (72/12/16 events) to prevent data leakage; (ii) fully fine-tuned the 300-million-parameter `Prithvi-EO-2.0-300M-TL-Sen1Floods11` model (ViT encoder, UperNet decoder) using TerraTorch on a free-tier Tesla T4 GPU; and (iii) evaluated both the published checkpoint and the fine-tuned model on 16 unseen flood events. Fine-tuning raised the flood-class IoU from 0.337 to 0.375 (+11.3%) and precision from 0.391 to 0.560 (+43.3%), while recall fell from 0.711 to 0.532, i.e. the model became markedly more conservative and produced far fewer false alarms.

In parallel we studied a mentor-provided benchmark on the global Sen1Floods11 dataset, in which a TerraMind-small foundation model (IoU 0.807) outperformed a from-scratch U-Net (0.765), a simple CNN (0.762) and Otsu thresholding (0.230), and in which the U-Net proved largely insensitive to the number of training labels. On the physics side we reviewed the one-dimensional Saint-Venant equations, implemented a MacCormack finite-difference solver that reproduces flood-wave attenuation ($400 \rightarrow 230\,\mathrm{m^3\,s^{-1}}$ over 20 km) and translation, performed a parameter-sensitivity study, and set up a physics-informed neural network (PINN) code base. We also designed an inference and decision-support workflow and conducted a field visit to Pawna Dam. The report closes with an analysis of limitations and a detailed plan for the second half of the semester, including threshold calibration, Sentinel-1 SAR fusion, FastFlood benchmarking, PINN validation and a Mahanadi-basin pilot.

Keywords: flood inundation mapping; geospatial foundation model; Prithvi-EO-2.0; Sentinel-2; NDEM; semantic segmentation; Saint-Venant equations; physics-informed neural networks; Mahanadi basin.

Contents

Certificate i

Declaration ii

Acknowledgements iii

Abstract iv

Abbreviations and Nomenclature x

1 Introduction 1

1.1 Background . . . 1
1.2 Industry context: ClimMaTech and the INCLINE/LSHMS platform . . . 1
1.3 Problem statement . . . 2
1.4 Objectives . . . 2
1.5 Scope of the mid-term report . . . 3
1.6 Organisation of the report . . . 3

2 Literature Review 4

2.1 Remote sensing of floods . . . 4
2.2 Classical thresholding . . . 4
2.3 Deep learning for flood segmentation . . . 4
2.4 Geospatial foundation models . . . 5
2.5 Physics-based flood routing . . . 5
2.6 Physics-informed neural networks . . . 6
2.7 Benchmarks for operational validation . . . 6
2.8 Research gap . . . 6

3 Study Region and Data 7

3.1 Study region . . . 7
3.2 Sentinel-2 multispectral imagery . . . 7
3.3 NDEM flood-extent maps (ground truth) . . . 7
3.4 Sen1Floods11 (benchmark dataset) . . . 8

4 Methodology 11

4.1 Overall workflow . . . 11
4.2 Dataset construction . . . 11
4.2.1 Pairing satellite scenes with flood maps . . . 11

4.2.2 Label design . . . 12
4.2.3 Tiling and storage optimisation . . . 12
4.2.4 Event-wise data split . . . 13
4.3 Model architecture . . . 13
4.4 Training configuration . . . 14
4.5 Evaluation methodology . . . 15
4.6 Computing environment . . . 16
5 Benchmark Study: Does a Foundation Model Beat a U-Net? 17
5.1 Purpose . . . 17
5.2 Experimental protocol . . . 17
5.3 Models compared . . . 18
5.4 Results . . . 18
5.4.1 Label efficiency . . . 19
5.4.2 Error analysis . . . 19
5.5 Implications for Prithvi Flood-IN . . . 20
6 Prithvi Flood-IN: Implementation and Results 21
6.1 Implementation . . . 21
6.2 Engineering challenges and resolutions . . . 21
6.3 Training outcome . . . 22
6.4 Results on the held-out test set . . . 22
6.5 Interpretation . . . 23
6.6 Pending analyses . . . 24
7 Physics-Based Flood Routing: Saint-Venant Equations and PINNs 25
7.1 Motivation . . . 25
7.2 Governing equations . . . 25
7.2.1 Approximations . . . 26
7.2.2 Applications . . . 26
7.3 Numerical solver developed in this project . . . 26
7.3.1 Results . . . 27
7.3.2 Parameter sensitivity . . . 27
7.4 Physics-informed neural network code base . . . 28
8 Inference Workflow, Applications and Field Visit 30
8.1 Inference workflow . . . 30
8.2 Applications . . . 31
8.3 Field visit: Pawna Dam . . . 32
8.4 Industry visit (proposed) . . . 32
9 Discussion and Limitations 34
9.1 Key findings so far . . . 34

9.2 Limitations . . . 34
10 Work Plan for the Second Half **35**
10.1 Milestones . . . 35
10.2 Planned technical work . . . 36
10.3 Expected deliverables . . . 36
10.4 Roles and responsibilities . . . 36
10.5 Risks . . . 37
11 Conclusions **38**
References **39**
A Code Listings **41**
A.1 Saint-Venant MacCormack solver (`sve_demo.py`) . . . 41
A.2 Masked loss for unlabelled pixels (from the benchmark notebook) . . . 42
A.3 Saint-Venant PDE residual for a PINN (illustrative sketch, TensorFlow 1.x) 42
B Project Log **43**
C Data and Code Locations **44**

List of Figures

3.1 Sen1Floods11 hand-labelled subset: chips per country, water fraction per chip by country, and distribution of unlabelled fraction. Source: mentor benchmark notebook [15]. . . . 9
3.2 Sample Sen1Floods11 chips: Sentinel-2 RGB, Sentinel-1 VV, Sentinel-1 VH and the hand label. Optical imagery is often cloud-obscured, while radar water is dark but easily confused with other dark surfaces. Source: [15]. . . 9
3.3 Distribution of Sentinel-1 VH backscatter over water and land pixels for the sample chips (median -19.1 dB vs. -15.3 dB). The overlap explains why a fixed threshold is insufficient. Source: [15]. . . . 10

4.1 Data-curation funnel from the initial event catalogue to the final dataset. . . 12
4.2 Pipeline for constructing training chips from Sentinel-2 scenes and NDEM flood maps. . . . 12
4.3 Distribution of events, scenes and chips across the three splits. . . . 13
4.4 Encoder–neck–decoder–head architecture of Prithvi Flood-IN. . . . 14

5.1 (a) IoU and F1 of the four methods on the Sen1Floods11 test split; (b) U-Net test IoU as a function of the number of labelled training chips, with TerraMind and Otsu as reference lines. Data from [15]. 19
5.2 Qualitative comparison on three Sen1Floods11 test chips: Sentinel-1 VH, truth, U-Net and TerraMind water probability, and U-Net error map (red: false alarm; blue: missed water). Source: [15]. 20

6.1 Training timeline: epochs run (dark), best checkpoint at epoch 7 (teal), and early stop at epoch 15 with patience 8. 22
6.2 Baseline versus fine-tuned metrics on the held-out Indian test set. 23
6.3 Baseline and fine-tuned operating points in precision–recall space with iso-F1 contours. 24

7.1 Simulated discharge hydrographs at the inflow, 8 km and 20 km. 27
7.2 Peak attenuation and travel time to 20 km for each scenario. 28
7.3 Structure of a physics-informed neural network for the Saint-Venant equations. 29

8.1 Inference and decision-support workflow. 30
8.2 Project team at Pawna Dam, Maval, Pune district, Maharashtra, during the monsoon **[date of visit]**. 32

10.1 Project timeline (Gantt chart). 35

List of Tables

3.1 Sentinel-2 bands used as model input, in channel order. 7
3.2 Products in the Sen1Floods11 v1.1 hand-labelled subset (all 512 × 512, EPSG:4326). 8
3.3 Sen1Floods11 official split by country. All 10 countries appear in more than one split. 8

4.1 Event-wise split of the Prithvi Flood-IN dataset. 13
4.2 Training configuration for Prithvi Flood-IN. 15
4.3 Software and hardware environment. 16

5.1 Methods in the benchmark study. 18
5.2 TerraMind fine-tuning configuration as executed in the benchmark notebook. 18
5.3 Benchmark results on the Sen1Floods11 test split (90 chips). Source: [15]. . 18
5.4 Error decomposition on chip Sri-Lanka_713926. Source: [15]. 19

6.1 Engineering problems encountered and their resolution. 21
6.2 Flood-class metrics on the held-out test set (16 events, 32 scenes, 737 chips). 22

7.1 Hierarchy of flood-routing approximations of the momentum equation. . . . 26
7.2 Parameters of the reference routing experiment (synthetic channel). 27
7.3 Sensitivity of peak discharge and travel time at 20 km to channel and flood parameters. 28
7.4 Workflow of the mentor-provided PINN code base. 29

8.1 Application pathways for Prithvi Flood-IN. 31

9.1 Limitations of the current work and planned mitigations. 34

10.1 Project milestones and status as of 5 October 2026. 35
10.2 Primary responsibilities of team members. 37
10.3 Project risks and mitigation. 37

Abbreviations and Nomenclature

Abbreviation	Meaning
AdamW	Adam optimiser with decoupled weight decay
AOI	Area of interest
BC / IC	Boundary condition / initial condition
CFL	Courant–Friedrichs–Lewy (stability) condition
CNN	Convolutional neural network
CWC	Central Water Commission
DEM	Digital elevation model
DUGC	Departmental Undergraduate Committee
EO	Earth observation
FN / FP / TP	False negative / false positive / true positive
GPU	Graphics processing unit
HIS	Hydrological information system
HLS	Harmonized Landsat and Sentinel-2
IDP	Integrated Design Project
IoU	Intersection over Union (Jaccard index)
MAE	Masked autoencoder
NDEM	National Database for Emergency Management (ISRO/NRSC)
NIR / SWIR	Near infrared / shortwave infrared
NRSC	National Remote Sensing Centre
PDE	Partial differential equation
PINN	Physics-informed neural network
SAR	Synthetic aperture radar
SVE	Saint-Venant equations
ViT	Vision Transformer

Symbol	Meaning (unit)
A	Cross-sectional flow area (m^2)
Q	Discharge ($\mathrm{m}^3\,\mathrm{s}^{-1}$)
h	Flow depth (m)
B	Channel width (m)
R	Hydraulic radius (m)
S_0, S_f	Bed slope, friction slope (–)
n	Manning roughness coefficient ($\mathrm{s}\,\mathrm{m}^{-1/3}$)
q	Lateral inflow per unit length ($\mathrm{m}^2\,\mathrm{s}^{-1}$)

g	Gravitational acceleration (9.81 $\mathrm{m\,s^{-2}}$)
Δx, Δt	Spatial and temporal step (m, s)

CHAPTER 1

Introduction

1.1 Background

Floods are the most frequent and most damaging natural hazard in India. Every south-west monsoon (June–September) brings riverine flooding to the Ganga, Brahmaputra, Mahanadi, Godavari and Krishna basins, while short-duration extreme rainfall causes flash and urban floods in hilly and densely built regions. The consequences — loss of life, displacement, damage to crops, roads, bridges and power infrastructure — recur annually, and climate change is expected to intensify extreme rainfall.

Effective response during a flood depends on knowing *where the water is*. Disaster-management authorities need inundation extents within hours of a satellite overpass to prioritise evacuation, route relief, identify cut-off villages and estimate crop and infrastructure losses. In India, official flood layers are disseminated by the National Database for Emergency Management (NDEM) of ISRO/NRSC. These layers are of high quality but are produced through analyst-supervised interpretation, which limits how quickly and how often they can be generated.

Two complementary families of tools are available to automate this task:

1. **Physics-based hydrodynamic models**, which solve the Saint-Venant (shallow-water) equations to predict how a flood wave propagates through a river network and onto the floodplain. Their accuracy is limited by the availability of river cross-sections, bed slope, roughness and elevation data.
2. **Data-driven Earth-observation (EO) models**, which classify satellite imagery pixel by pixel into flooded and non-flooded land. Recent *geospatial foundation models*, pre-trained on millions of unlabelled satellite images, promise strong performance from comparatively few labelled examples.

This project brings the two together in the context of an industry product.

1.2 Industry context: ClimMaTech and the INCLINE/LSHMS platform

ClimMaTech Pvt. Ltd., an IIT Mandi-incubated company, together with Prof. Basudev Biswal's group at IIT Bombay, is developing a hydrological information and flood-modelling platform (referred to internally as INCLINE/LSHMS). At the project kick-off (3 August 2026)

the team was introduced to the existing software, and the following characteristics and gaps were identified:

- The platform uses two families of input: *topographic data* (river geometry and elevation) and *rainfall data* (observed records for two pilot regions, and future rainfall for identifying flood-prone areas).
- Field surveys of river geometry are expensive — of the order of **Rs. 1 lakh for 10–15 km** of river — so river width is taken from observations while depth is estimated through empirical relations.
- Available elevation data carry a vertical uncertainty of about $\pm$**1 m**, and slopes are derived through a smoothing algorithm that offers scope for improvement.
- The platform already generates automated, LLM-written flood reports in 16 languages.

A satellite-derived, observed flood extent is therefore valuable to the platform in two ways: directly, as an inundation product for decision support, and indirectly, as an independent observation against which physically based simulations can be checked and calibrated.

1.3 Problem statement

Prithvi Flood-IN: Pre- and During-Event Inundation Mapping with Prithvi-EO-2.0 over the Indian Subcontinent.

The publicly released flood-segmentation version of IBM–NASA's Prithvi-EO-2.0 foundation model was fine-tuned on the global Sen1Floods11 dataset, which contains only 68 Indian chips out of 446. Indian flood scenes differ systematically from that training distribution: rivers carry high sediment loads, monsoon cloud cover is persistent, and land cover ranges from irrigated paddy to dense urban areas. Applied to Indian events, the published model detects most flooded area but produces many false alarms (Chapter 6). The problem addressed is to *adapt* this foundation model to Indian conditions using official Indian flood maps, to evaluate it rigorously on unseen events, and to define how its output can be integrated with hydrological data and physics-based reasoning for flood response in the Mahanadi basin.

1.4 Objectives

The aim of the project is to contribute towards the product development of the ClimMaTech–IIT Bombay team. The specific objectives are:

O1. Develop flood inundation mapping with machine-learning techniques using available satellite images.

O2. Build an India-specific flood training dataset by pairing Sentinel-2 imagery with NDEM flood-extent maps from real Indian flood events, with rigorous quality control.

O3. Fine-tune the Prithvi-EO-2.0 foundation model for Indian conditions and quantify its improvement over the published model on unseen events.

O4. Benchmark foundation models against classical and from-scratch deep-learning baselines.

O5. Develop an inference workflow that produces georeferenced flood-extent maps and design their integration with dam, gauge and infrastructure layers.

O6. Study physics-based flood routing through the Saint-Venant equations and physics-informed neural networks, and link satellite observations to physics-based model validation.

O7. Pilot the system on the Mahanadi basin and prototype a decision-support dashboard.

[Industry mentors to confirm or add specific objectives]

1.5 Scope of the mid-term report

This report documents work completed between 29 July and 5 October 2026: the literature and platform study, dataset curation, baseline evaluation and full fine-tuning of Prithvi-EO-2.0, the benchmark study on Sen1Floods11, the Saint-Venant and PINN study, the inference and application design, and a field visit. Objectives O1–O4 are substantially complete; O5 and O6 are in progress; O7 is planned for the second half of the semester.

1.6 Organisation of the report

Chapter 2 reviews the literature. Chapter 3 describes the study region and data. Chapter 4 presents the methodology for dataset construction, model architecture, training and evaluation. Chapter 5 reports the benchmark study on Sen1Floods11. Chapter 6 presents the implementation and results of Prithvi Flood-IN. Chapter 7 covers the Saint-Venant equations, our numerical experiments and the PINN code base. Chapter 8 describes the inference workflow, applications and the field visit. Chapter 9 discusses findings and limitations, Chapter 10 lays out the plan for the remainder of the project, and Chapter 11 concludes.

CHAPTER 2

Literature Review

2.1 Remote sensing of floods

Optical sensors such as the Multispectral Instrument on Sentinel-2 measure reflected sunlight in 13 spectral bands at 10–60 m resolution with a revisit of about five days. Open water absorbs strongly in the near-infrared (NIR) and shortwave-infrared (SWIR) bands, so spectral indices such as the Normalised Difference Water Index, $\mathrm{NDWI} = (G - \mathrm{NIR})/(G + \mathrm{NIR})$, and its modified form using SWIR, $\mathrm{MNDWI} = (G - \mathrm{SWIR1})/(G + \mathrm{SWIR1})$, have been used for decades to delineate water. The principal limitation of optical sensing during floods is cloud cover, which is especially persistent during the Indian monsoon.

Synthetic aperture radar (SAR), such as the C-band Sentinel-1, is an active sensor that penetrates cloud and works day and night. Calm water acts as a specular reflector and appears dark (typically below about -15 dB in VV and -22 dB in VH polarisation). SAR has well-known failure modes: smooth dry surfaces (tarmac, sand) and radar shadow are also dark, producing false alarms; flooded vegetation produces bright double-bounce returns, producing misses; wind roughens water surfaces; and urban areas scatter strongly. Optical and radar data therefore fail in different situations, which motivates multi-sensor approaches.

2.2 Classical thresholding

The most widely used operational method for SAR flood mapping is histogram thresholding, in particular Otsu's method [8], which chooses the grey-level threshold that maximises the between-class variance of a bimodal histogram. It needs no training data, but its performance degrades when water occupies a small fraction of the scene (the histogram is not bimodal) or when land and water backscatter overlap. The mentor benchmark (Chapter 5) found that 50.3% of water pixels in sample Sen1Floods11 chips were brighter in VH than the darkest 5% of land pixels, which illustrates why a single threshold is insufficient.

2.3 Deep learning for flood segmentation

Convolutional neural networks (CNNs) learn spatial and spectral features directly from labelled data. The U-Net [9] — an encoder–decoder with skip connections — has been the default architecture for semantic segmentation since 2015 and remains a strong baseline. The Sen1Floods11 dataset [1] provides 446 hand-labelled 512×512 chips at 10 m resolu-

tion from 11 flood events across six continents, with co-registered Sentinel-1, Sentinel-2, DEM and JRC permanent-water layers, and has become the standard benchmark for flood segmentation. Its official splits share events between training and test sets, so reported scores measure generalisation *within* seen events rather than transfer to new floods.

2.4 Geospatial foundation models

A foundation model is a large network pre-trained on a very large unlabelled corpus with a self-supervised objective, typically reconstructing masked portions of its input, and then adapted to downstream tasks with relatively few labels.

- **Prithvi** [3] was released by IBM and NASA as a Vision Transformer (ViT) masked autoencoder pre-trained on Harmonized Landsat–Sentinel-2 (HLS) imagery over the United States.
- **Prithvi-EO-2.0** [2] extends pre-training to a global, multi-temporal HLS corpus and releases 300 M and 600 M parameter variants, together with task-specific checkpoints including `Prithvi-EO-2.0-300M-TL-Sen1Floods11`, which is the starting point of this project. It uses six HLS bands: Blue, Green, Red, narrow NIR, SWIR-1 and SWIR-2.
- **TerraMind** [4] is a multimodal, generative foundation model pre-trained jointly on Sentinel-2 L1C/L2A, Sentinel-1 GRD/RTC, DEM, NDVI, land cover and RGB, released in tiny (6.1 M), small (22.8 M), base (87.9 M) and large (305.9 M) sizes.
- **TerraTorch** [5] is IBM's open-source fine-tuning toolkit built on PyTorch Lightning, which attaches task heads to pre-trained encoders, handles multiband GeoTIFF loading and normalisation, and expresses complete training runs as reproducible YAML configurations.

Whether foundation models actually outperform well-tuned supervised baselines is contested: the PANGAEA benchmark [6] evaluated twelve geospatial foundation models and found that a plain supervised U-Net outperformed all of them on Sen1Floods11. Results are also highly sensitive to hyper-parameters; variations of up to 51 percentage points in mIoU across learning rates have been reported [7]. These findings motivated the benchmark study in Chapter 5.

2.5 Physics-based flood routing

Unsteady open-channel flow is governed by the one-dimensional Saint-Venant equations [12], expressing conservation of mass and momentum. Depending on which momentum terms are retained, routing models are classified as kinematic-wave, diffusive-wave or full dynamic-wave models. Operational hydrodynamic packages such as HEC-RAS and LISFLOOD-

FP solve these equations in one or two dimensions to predict stage, discharge and inundation extent. Their accuracy depends on channel geometry, roughness and boundary conditions — precisely the inputs identified as costly and uncertain for the ClimMaTech platform.

2.6 Physics-informed neural networks

Physics-informed neural networks (PINNs) [13] train a neural network to approximate the solution of a PDE by minimising a loss that combines misfit to observations with the residual of the governing equations, evaluated through automatic differentiation at collocation points. For river hydraulics, PINNs can assimilate sparse gauge observations while respecting the Saint-Venant equations, offering a route to estimating flow states and uncertain parameters between gauges.

2.7 Benchmarks for operational validation

The FastFlood validation framework, shared by the teaching assistant during the project, benchmarks flood simulations against more than 250 observed satellite flood extents, 450 discharge gauges and about 2,000 virtual stations, and compares AI-corrected rainfall against over 10,000 stations worldwide. It provides a useful external reference for the second phase of this project.

2.8 Research gap

From the review, three gaps directly relevant to this project emerge:

1. Published foundation-model flood checkpoints are trained and evaluated predominantly on non-Indian data, with evaluation splits that share events between training and testing.
2. There is no openly documented, India-specific, event-wise-split training set derived from official NDEM flood maps.
3. Satellite-observed flood extents and physics-based routing are rarely combined within a single operational information system for Indian river basins.

This project addresses the first two gaps in its first half and is designed to address the third in its second half.

CHAPTER 3

Study Region and Data

3.1 Study region

The geographic scope of the dataset is the Indian subcontinent, using every NDEM flood layer for which matching satellite imagery could be found. The application pilot is the **Mahanadi basin**, one of India's major east-flowing river systems, draining parts of Chhattisgarh and Odisha into the Bay of Bengal. The basin is regulated by the Hirakud reservoir and is prone to monsoon floods in its deltaic reaches, making it a natural test-bed for combining reservoir operations, river gauging and satellite flood mapping.

3.2 Sentinel-2 multispectral imagery

Input imagery comes from the Copernicus Sentinel-2 mission (ESA), which is freely available. Six bands, in the order expected by Prithvi-EO-2.0, are used (Table 3.1). The two SWIR bands carry the strongest water signal because water absorbs strongly at these wavelengths.

Table 3.1: Sentinel-2 bands used as model input, in channel order.

Ch.	Band	Sentinel-2 band	Central wavelength (nm)	Role in flood mapping
1	Blue	B02	490	Turbidity, atmospheric effects
2	Green	B03	560	Water reflectance peak; NDWI
3	Red	B04	665	Vegetation and sediment contrast
4	NIR (narrow)	B8A	865	Strong water absorption; vegetation
5	SWIR-1	B11	1610	Very strong water absorption; MNDWI
6	SWIR-2	B12	2190	Very strong water absorption; burn/soil

3.3 NDEM flood-extent maps (ground truth)

Ground-truth labels are the official flood-inundation layers published by NDEM (ISRO/NRSC) for real Indian flood events from 2021 to 2026. NDEM layers mark *newly inundated* land for a given date; they do not mark permanent rivers and lakes as flooded. This property matters for label design (Section 4.2.2).

3.4 Sen1Floods11 (benchmark dataset)

For the benchmark study, the global Sen1Floods11 v1.1 hand-labelled subset [1] was used (1.8 GB). Table 3.2 summarises its contents and Table 3.3 its official splits as inspected in the mentor notebook.

Table 3.2: Products in the Sen1Floods11 v1.1 hand-labelled subset (all 512×512, EPSG:4326).

Product	Files	Bands	Content
S1GRDHand	446	2	Sentinel-1 VV, VH backscatter (dB, float32)
S2L1CHand	446	13	Sentinel-2 L1C top-of-atmosphere reflectance (int16)
LabelHand	446	1	Hand label: 1 water, 0 dry, -1 undetermined
S1OtsuLabelHand	446	1	Otsu-threshold label
JRCWaterHand	446	1	JRC permanent-water layer
CopernicusDEM	446	1	Elevation (int16)

Table 3.3: Sen1Floods11 official split by country. All 10 countries appear in more than one split.

Country	Train	Val	Test	Total
USA	41	14	14	69
India	40	14	14	68
Paraguay	39	14	14	67
Ghana	31	11	11	53
Sri Lanka	24	9	9	42
Mekong	18	6	6	30
Spain	18	6	6	30
Pakistan	16	6	6	28
Somalia	15	5	6	26
Nigeria	10	4	4	18
Total	**252**	**89**	**90**	**431**

An exploratory scan of 431 label files showed a mean water fraction of 0.107 per chip, 145 chips with less than 1% water, 26 chips with more than 50% water, and a mean unlabelled (-1) fraction of 0.132. Class balance therefore varies enormously from chip to chip (Figure 3.1), and a significant share of pixels must be excluded from both the loss and the metrics.

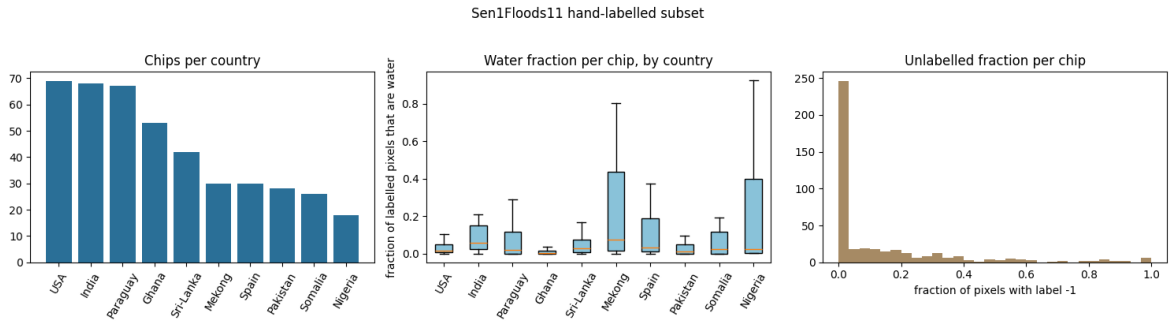


Figure 3.1: Sen1Floods11 hand-labelled subset: chips per country, water fraction per chip by country, and distribution of unlabelled fraction. Source: mentor benchmark notebook [15].

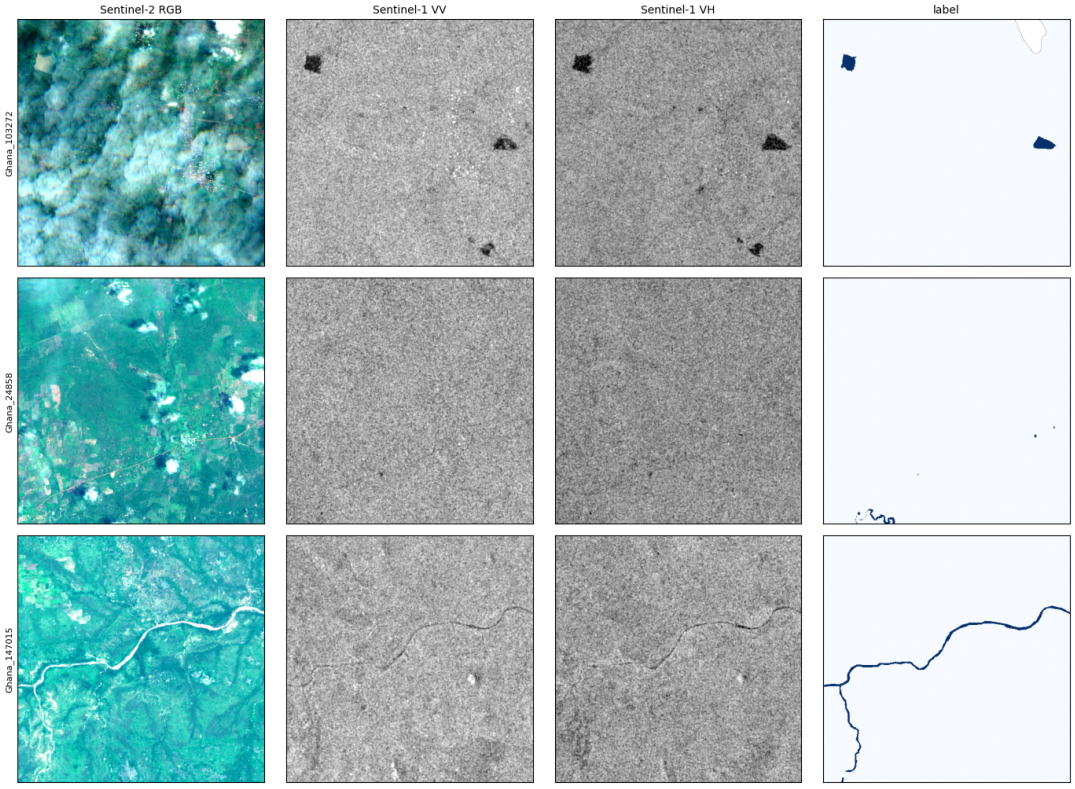


Figure 3.2: Sample Sen1Floods11 chips: Sentinel-2 RGB, Sentinel-1 VV, Sentinel-1 VH and the hand label. Optical imagery is often cloud-obscured, while radar water is dark but easily confused with other dark surfaces. Source: [15].

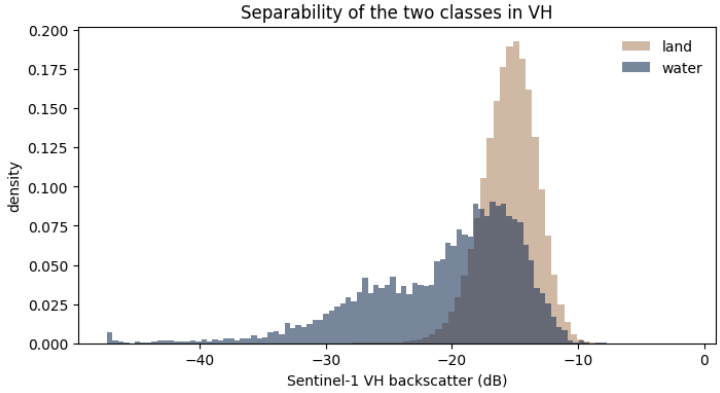


Figure 3.3: Distribution of Sentinel-1 VH backscatter over water and land pixels for the sample chips (median -19.1 dB vs. -15.3 dB). The overlap explains why a fixed threshold is insufficient. Source: [15].

CHAPTER 4

Methodology

4.1 Overall workflow

The methodology has four connected components: (i) construction of an India-specific labelled dataset; (ii) adaptation of the Prithvi-EO-2.0 foundation model by full fine-tuning; (iii) fair, leakage-free evaluation against the published model; and (iv) an inference workflow that converts model output into decision-support products. A parallel physics track (Chapter 7) studies flood routing.

4.2 Dataset construction

4.2.1 Pairing satellite scenes with flood maps

For each NDEM flood layer, a Sentinel-2 scene was sought that satisfied three conditions: (a) acquisition *during* the mapped flood, close to the NDEM map date; (b) spatial overlap with the mapped flood area; and (c) sufficiently low cloud cover over the flooded region. An automated pipeline searched candidate scenes, checked cloud cover, verified that the flood outline overlapped visibly wet pixels, and selected the best scene per event. A sample of matches was further verified visually by overlaying the NDEM outline on the false-colour image.

The initial project spreadsheet listed 221 flood events, of which only 5 were directly usable. Extending the search to all 1,022 dated flood-extent layers on the NDEM server yielded **180 usable scene–label pairs covering 100 distinct flood events** (Figure 4.1). The main reasons for rejection were absence of imagery on suitable dates, cloud cover over the flooded area, and insufficient overlap between the scene and the mapped flood.

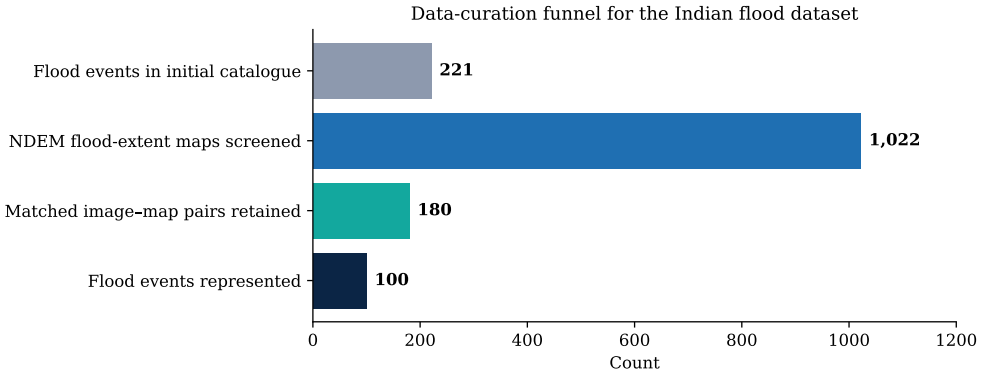


Figure 4.1: Data-curation funnel from the initial event catalogue to the final dataset.

4.2.2 Label design

Each pixel of the target raster takes one of three values:

- **1 — flood**: inside the NDEM inundation polygon;
- **0 — dry land**: inside the scene and outside the inundation polygon;
- -1 **— ignore**: cloud, cloud shadow, permanent water (rivers, lakes, reservoirs) or area outside the officially mapped flood zone.

Assigning permanent water to the ignore class is essential. Because NDEM marks only newly flooded land, labelling the river channel as "dry" would teach the model that open water is dry land. Similarly, cloud-covered pixels cannot be verified and are excluded rather than guessed. The ignore value is passed to the loss as `ignore_index = -1`, and the same mask is applied when computing every metric. The construction pipeline is summarised in Figure 4.2.

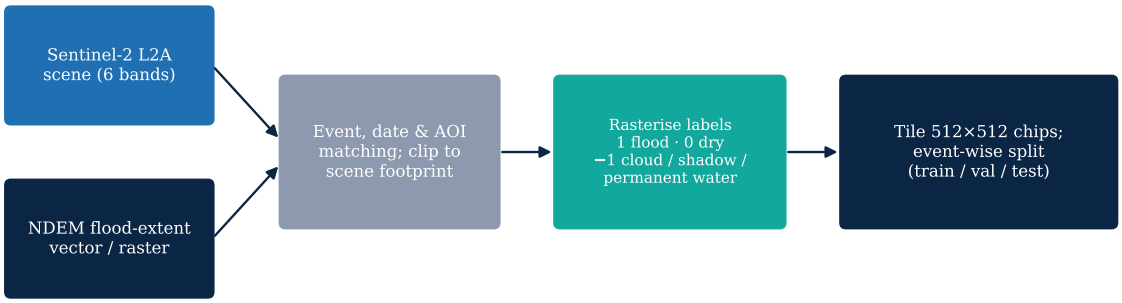


Figure 4.2: Pipeline for constructing training chips from Sentinel-2 scenes and NDEM flood maps.

4.2.3 Tiling and storage optimisation

Scenes and label rasters were tiled into chips of 512×512 pixels **[confirm ground sampling distance; project notes state approx. 15 km $\times$ 15 km per chip]**. This gave **3,584 chips**, of

which 2,401 contain flooded pixels and 1,183 do not. Several NDEM files were very large (over 150 MB) because they covered regions far bigger than the matching scene; clipping each label to its scene footprint reduced the dataset from about **1.9 GB to 625 MB**.

4.2.4 Event-wise data split

Two scenes from the same flood are highly correlated. If one were used for training and the other for testing, the test score would reward recognising the scene rather than understanding floods. The split was therefore performed by *whole flood event* (Table 4.1, Figure 4.3). This is a stricter protocol than the official Sen1Floods11 split, in which every country appears in more than one subset.

Table 4.1: Event-wise split of the Prithvi Flood-IN dataset.

Split	Events	Scenes	Chips	Purpose
Train	72	130	2,635	Weight updates
Validation	12	18	212	Model selection after every epoch
Test	16	32	737	Held out; used once for final scoring
Total	**100**	**180**	**3,584**	

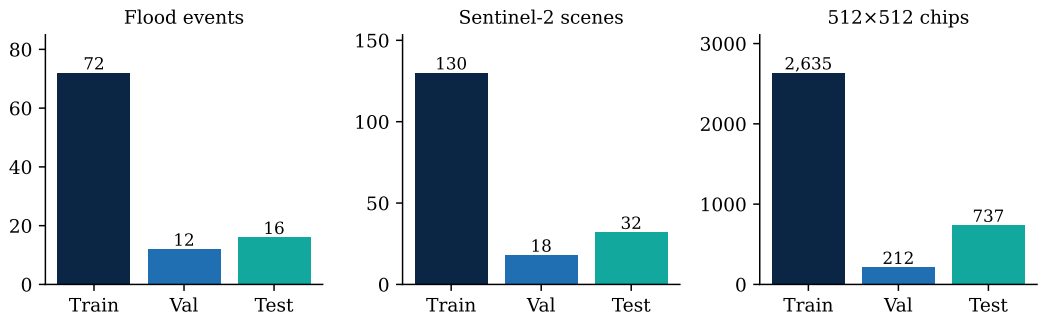


Figure 4.3: Distribution of events, scenes and chips across the three splits.

4.3 Model architecture

We reuse the exact architecture and starting weights of the published checkpoint `ibm-nasa-geospatial/Prithvi-EO-2.0-300M-TL-Sen1Floods11`, built with TerraTorch's `EncoderDecoderFactory` inside a `SemanticSegmentationTask` (Figure 4.4).

- **Encoder** (`prithvi_eo_v2_300_tl`): a ViT-Large encoder of about 300 M parameters (24 transformer blocks) pre-trained as a masked autoencoder on global HLS data, already transfer-learned once on Sen1Floods11. The image is divided into patches, each embedded as a token; self-attention models relations between all patches.

- **Neck**: (1) `SelectIndices [5, 11, 17, 23]` taps four evenly spaced encoder depths, combining shallow detail and deep semantics; (2) `ReshapeTokensToImage` folds each token sequence back into a 2-D feature grid; (3) `LearnedInterpolateToPyramidal` learns upsampling to four spatial scales, producing a feature pyramid.
- **Decoder**: `UperNetDecoder` with 256 channels, which fuses the multi-scale features (pyramid pooling and feature pyramid) into a single full-resolution map.
- **Head**: a per-pixel classifier producing two logits (not-flood, flood), with dropout 0.1.

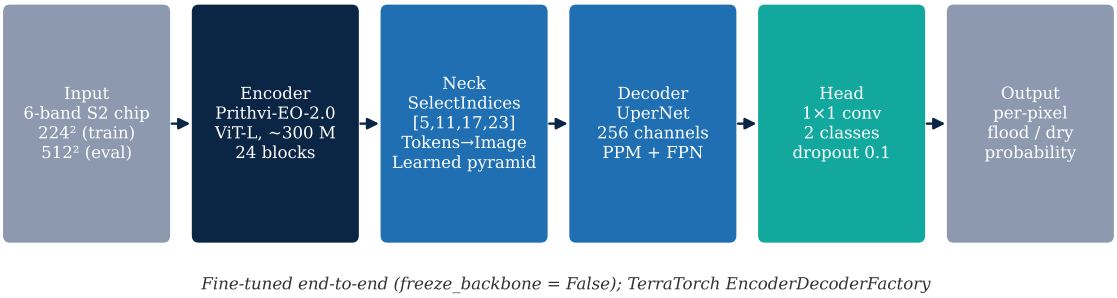


Figure 4.4: Encoder–neck–decoder–head architecture of Prithvi Flood-IN.

4.4 Training configuration

Training uses the per-pixel cross-entropy loss over valid pixels,

$$\mathscr{L} = -\frac{1}{|\Omega|} \sum_{i \in \Omega} \sum_{c \in \{0,1\}} y_{i,c} \log \hat{p}_{i,c}, \qquad \Omega = \{\, i : y_i \neq -1 \,\}, \tag{4.1}$$

where $\hat{p}_{i,c}$ is the softmax probability of class c at pixel i. Table 4.2 lists all settings.

Table 4.2: Training configuration for Prithvi Flood-IN.

Setting	Value	Rationale
Loss	Cross-entropy	Standard per-pixel classification loss
`ignore_index`	-1	Cloud, shadow, permanent water and unmapped pixels excluded
Optimiser	AdamW	Adaptive optimiser with decoupled weight decay
Learning rate	5×10^{-5}	Small, to refine rather than overwrite pre-trained knowledge
Weight decay	0.05	Regularisation against overfitting
Scheduler	Cosine annealing	Smooth decay of learning rate
`freeze_backbone`	False	Full fine-tune of the 300 M-parameter encoder
`freeze_decoder`	False	Decoder also adapted
Batch size	8	Fits T4 memory with mixed precision
Training crop	224×224 (random)	Memory and speed; full 512×512 at evaluation
Precision	16-bit mixed	Lower memory, faster training
Maximum epochs	20	Planned budget
Early stopping	Patience 8 on val. flood IoU	Prevents overfitting and wasted compute
Checkpoint selection	Best `val/multiclass jaccardindex_flood`	Most generalisable epoch, not the last

4.5 Evaluation methodology

For each test chip, predictions are compared with labels over the valid pixel set Ω. Accumulating true positives (TP), false positives (FP) and false negatives (FN) for the flood class over all 737 test chips, the metrics are

$$\text{Precision} = \frac{TP}{TP+FP}, \qquad \text{Recall} = \frac{TP}{TP+FN}, \tag{4.2}$$

$$\text{IoU} = \frac{TP}{TP+FP+FN}, \qquad F_1 = \frac{2TP}{2TP+FP+FN} = \frac{2PR}{P+R}. \tag{4.3}$$

IoU is the primary metric because, unlike pixel accuracy, it cannot be inflated by predicting "no flood" everywhere in scenes dominated by dry land. The two are related by $F_1 = 2\,\text{IoU}/(1+\text{IoU})$, which we use as a consistency check. Both the published baseline and the fine-tuned model are scored with identical code, identical metric definitions and the identical test set, so the only variable is the model weights. In addition, the following analyses are prepared: confusion matrices; per-event metrics; inference time per chip; and four-panel visual comparisons (Sentinel-2 image, NDEM truth, prediction, error map with hits, false alarms and misses).

4.6 Computing environment

Table 4.3: Software and hardware environment.

Component	Detail
Compute	Google Colab (free tier), NVIDIA Tesla T4, 15,360 MiB
System RAM	12.7 GB (shared virtual machine)
Peak GPU memory used	7.11 GiB
Python	3.13
Libraries	`torch` 2.6.0, `torchvision` 0.21.0, `torchaudio` 2.6.0, `terratorch` 0.99.8, `lightning` 2.5.0, `rasterio` 1.4.3
Training loop	PyTorch Lightning `Trainer`
Storage	Google Drive (persistence); local VM disk (active I/O)

CHAPTER 5

Benchmark Study: Does a Foundation Model Beat a U-Net?

5.1 Purpose

A foundation model is more expensive to run and harder to configure than a conventional network; it is worth adopting only if it outperforms a well-built baseline. Before committing to Prithvi-EO-2.0 for the Indian dataset, the team studied a benchmark notebook prepared by the industry mentors [15], which compares a classical threshold, two from-scratch CNNs and the TerraMind foundation model on Sen1Floods11 under a single, fair protocol. The results quoted below are those recorded in the notebook run on a Colab T4 GPU (15 September 2026) **[state whether the team re-ran it]**.

5.2 Experimental protocol

- **Data:** Sen1Floods11 official splits — 252 training, 89 validation and 90 test chips.
- **Inputs:** 15 channels (13 Sentinel-2 L1C bands followed by Sentinel-1 VV and VH), standardised with TerraMind's pre-training mean and standard deviation for every model; radar no-data values below -80 dB replaced by the VV mean.
- **Metric:** water-class IoU and F1 over labelled pixels only, using a 0.5 probability cut-off, with identical scoring code for all methods.
- **Loss masking:** pixels labelled -1 excluded from the loss through a binary mask (`counts_in_loss`); the notebook stresses that omitting this step inflates every reported score.

5.3 Models compared

Table 5.1: Methods in the benchmark study.

Method	Parameters	Description
Otsu threshold	0	Per-chip Otsu threshold on Sentinel-1 VH; no training
Simple CNN	14,017	Three 3×3 conv layers with batch norm and ReLU; 7×7 receptive field; no downsampling
U-Net	119,409	Two-level encoder–decoder (width 16), max-pooling, transposed-conv upsampling and skip connections
TerraMind-S	≈22.8 M (backbone)	TerraMind v1 small, multimodal input {S2L1C, S1GRD}; necks SelectIndices [2, 5, 8, 11], ReshapeTokensToImage, LearnedInterpolateToPyramidal; U-Net decoder [256, 128, 64, 32]; 2-class head with 10% dropout

The CNN and U-Net were trained for 10 epochs with AdamW (learning rate 10^{-3}, batch size 4) and masked binary cross-entropy. TerraMind was fine-tuned with TerraTorch from a modified version of the official IBM configuration (`terramind_v1_tiny_sen1floods11.yaml`) with the settings in Table 5.2; the checkpoint used for scoring was `epoch=29-step=930.ckpt`.

Table 5.2: TerraMind fine-tuning configuration as executed in the benchmark notebook.

Setting	Value	Setting	Value
Backbone	`terramind_v1_small`	Max epochs	30
Backbone frozen	False	Learning rate	10^{-4}
Batch size	8	Precision	16-mixed
Data module	`GenericMultiModalDataModule`	Seed	42

5.4 Results

Table 5.3: Benchmark results on the Sen1Floods11 test split (90 chips). Source: [15].

Method	Training	Water IoU	Water F1
Otsu threshold (VH)	None	0.230	0.374
Simple CNN	∼2 min, 10 epochs	0.762	0.865
U-Net	∼4 min, 10 epochs	0.765	0.867
TerraMind-S (fine-tuned)	30 epochs	**0.807**	**0.893**

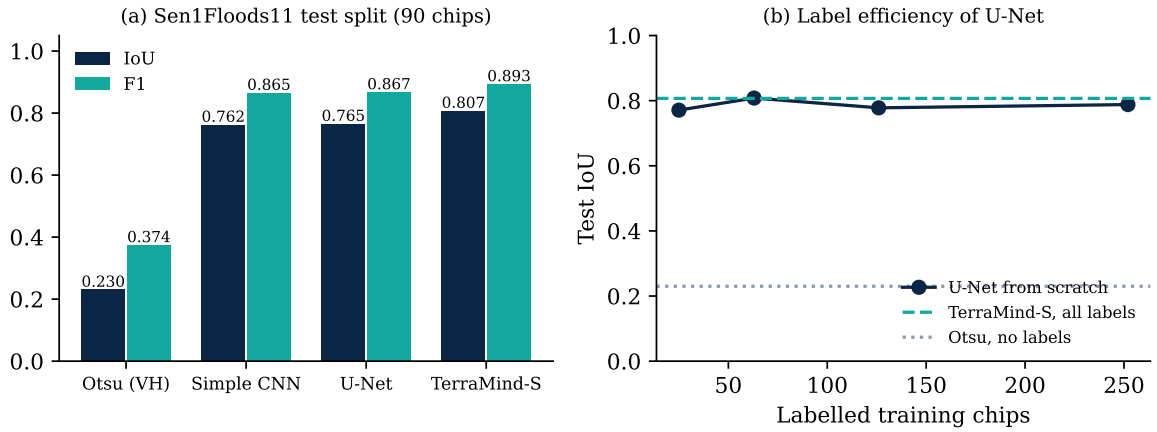


Figure 5.1: (a) IoU and F1 of the four methods on the Sen1Floods11 test split; (b) U-Net test IoU as a function of the number of labelled training chips, with TerraMind and Otsu as reference lines. Data from [15].

5.4.1 Label efficiency

Retraining the U-Net on 10%, 25%, 50% and 100% of the training chips gave IoU values of 0.771 (25 chips), 0.808 (63), 0.778 (126) and 0.788 (252). Going from 25 to 252 chips changed IoU by only +0.017. The curve is essentially flat, which indicates that on Sen1Floods11 the number of labels is not the bottleneck. The 100% value (0.788) differs from the main U-Net run (0.765) despite identical data and seed, which indicates run-to-run variability of about ± 0.02 IoU on this small test set.

5.4.2 Error analysis

Figure 5.2 shows test chips with radar input, truth, U-Net and TerraMind probabilities and the U-Net error map. Pixel counts on the chip `Sri-Lanka_713926` (Table 5.4) show that a better IoU does not imply fewer errors of both kinds: TerraMind finds the most water but also raises the most false alarms.

Table 5.4: Error decomposition on chip Sri-Lanka_713926. Source: [15].

Method	False alarms	Missed water	Correct water
U-Net	302	1,644	78,862
Simple CNN	419	1,078	79,428
TerraMind-S	1,339	619	79,887

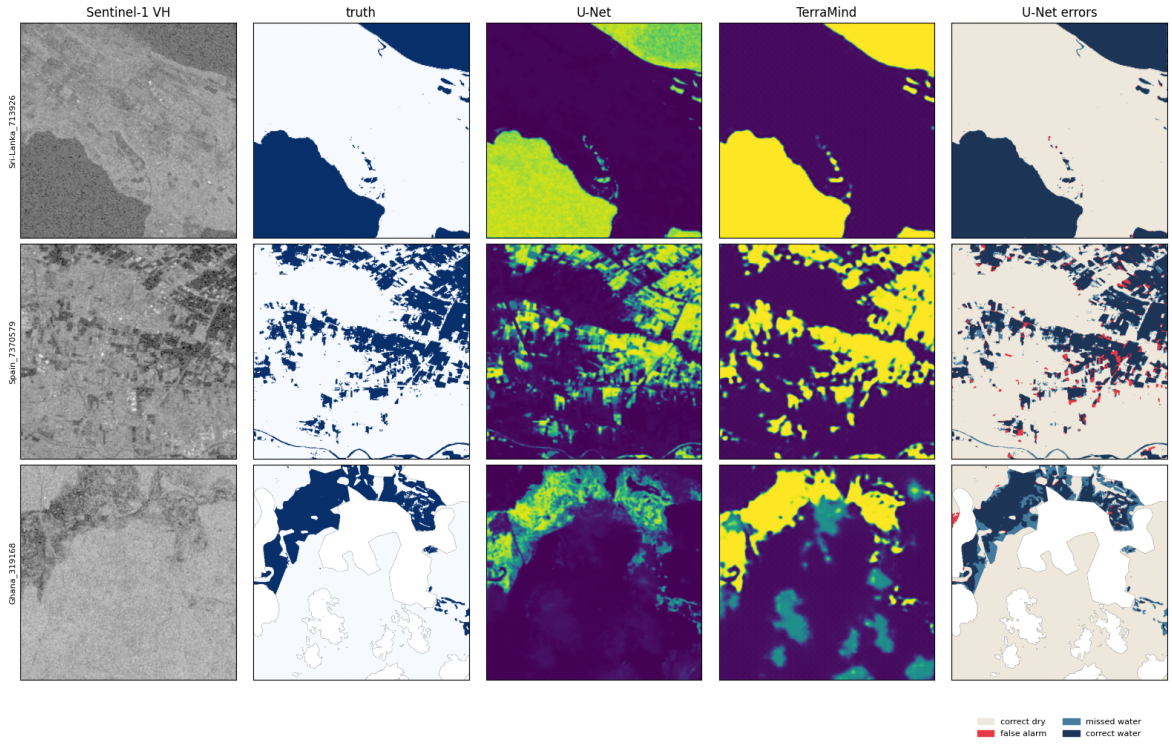


Figure 5.2: Qualitative comparison on three Sen1Floods11 test chips: Sentinel-1 VH, truth, U-Net and TerraMind water probability, and U-Net error map (red: false alarm; blue: missed water). Source: [15].

The notebook also converts a binary map into flooded area, the quantity an emergency manager actually asks for. On chip `Mekong_424793`, the U-Net at a cut-off of 0.5 predicted 0.22 km^2 against a true 0.40 km^2 (IoU 0.474), and the estimated area changed substantially when the cut-off was moved between 0.3 and 0.7. This motivates the threshold-calibration step planned for Prithvi Flood-IN.

5.5 Implications for Prithvi Flood-IN

1. With radar and optical inputs and within-event splits, the foundation model gave a modest but real gain (+0.04 IoU) over a from-scratch U-Net, consistent with the contested picture in the literature [6].
2. Absolute scores on Sen1Floods11 (IoU ≈ 0.8) are not comparable with our event-wise Indian test set, where every test flood is unseen; we expect, and observe, much lower absolute IoU.
3. Masking of unlabelled pixels, fixed-threshold metrics and identical scoring code are non-negotiable for a fair comparison. All three practices are adopted in our evaluation.
4. A from-scratch U-Net baseline on the Indian dataset is a necessary addition in the second half (Chapter 10).

CHAPTER 6

Prithvi Flood-IN: Implementation and Results

6.1 Implementation

The project code base is organised into a data module (`floodin/data.py`), a model factory (`floodin/model.py`), a fine-tuning script (`scripts/finetune.py`) and an evaluation script (`scripts/evaluate.py`). The data module reads the six-band chips and label rasters, applies the scaling and per-band standardisation expected by the pre-trained encoder, takes random 224×224 crops during training and full 512×512 chips at evaluation. Training was run from a Colab notebook (`colab_finetune.ipynb`) with the configuration in Table 4.2; checkpoints, logs and result files were stored on Google Drive.

6.2 Engineering challenges and resolutions

Running a 300 M-parameter model on free-tier infrastructure exposed several practical problems. Each was diagnosed by inspecting the running virtual machine (process list, memory use, disk I/O) rather than relying only on the training log, and training was always resumed from the last verified checkpoint.

Table 6.1: Engineering problems encountered and their resolution.

Problem	Symptom / cause	Resolution
Library version mismatch	Pre-installed `torchaudio` incompatible with pinned `torch`; crash on import	Pinned `torchaudio` in `requirements.txt`
Input scaling mismatch	Reflectance not on the scale used in pre-training; baseline model nearly useless	Rescaled inputs to the pre-training convention and applied checkpoint normalisation
Validation plotting crash	Default `plot_on_val` called a dataset interface not implemented	Set `plot_on_val=False`
Orphaned training process	After a Colab disconnect, a stale process kept writing to and corrupting the checkpoint on Drive	Terminated the process; restored last intact checkpoint; write locally first, then sync to Drive
Silent RAM exhaustion	Data-loader workers gradually consumed 12 GB RAM; OS killed training without error	`num_workers=0`, trading speed for stability

6.3 Training outcome

Training ran for 15 of the planned 20 epochs, after which early stopping triggered: no epoch from 8 to 15 improved on the best validation flood IoU of **0.2989**, reached at **epoch 7** (Figure 6.1). The epoch-7 checkpoint (`run1/checkpoints/best-07.ckpt`) was used for all test results. Peak GPU memory was 7.11 GiB.

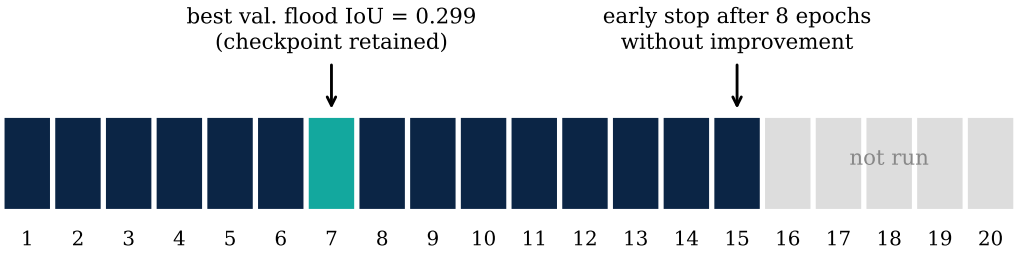


Figure 6.1: Training timeline: epochs run (dark), best checkpoint at epoch 7 (teal), and early stop at epoch 15 with patience 8.

6.4 Results on the held-out test set

Table 6.2 and Figure 6.2 compare the published Sen1Floods11 checkpoint (no Indian fine-tuning) with Prithvi Flood-IN on the 737 test chips from 16 unseen events. F1 is computed from precision and recall and agrees with the IoU through $F_1 = 2\,\mathrm{IoU}/(1+\mathrm{IoU})$.

Table 6.2: Flood-class metrics on the held-out test set (16 events, 32 scenes, 737 chips).

Metric	Baseline (published)	Prithvi Flood-IN	Absolute change	Relative change
IoU	0.3372	**0.3752**	+0.0380	**+11.3%**
Precision	0.3909	**0.5601**	+0.1692	**+43.3%**
Recall	**0.7106**	0.5324	−0.1782	−25.1%
F1-score†	0.5045	**0.5459**	+0.0414	+8.2%

† Derived as $2PR/(P+R)$ from the reported precision and recall.

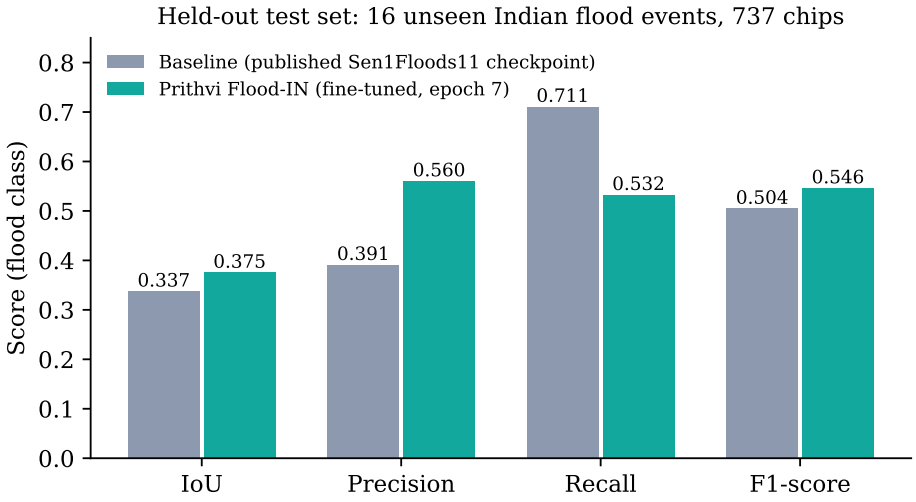


Figure 6.2: Baseline versus fine-tuned metrics on the held-out Indian test set.

6.5 Interpretation

- **The published model was over-eager.** It detected 71% of truly flooded pixels but only 39% of its flood predictions were correct, i.e. roughly three of every five "flood" pixels were false alarms on Indian scenes.
- **Fine-tuning recalibrated the model to Indian conditions.** Precision rose to 56%, cutting the false-alarm share from 61% to 44% of flood predictions.
- **The primary metric improved.** IoU rose by 11.3% and F1 by 8.2%, confirming a net improvement rather than a mere shift in operating point.
- **There is a precision–recall trade-off.** Recall fell from 0.71 to 0.53: the model became more conservative. Figure 6.3 shows that the fine-tuned model moved to a higher iso-F1 contour while shifting towards precision. Which operating point is preferable depends on the use: early warning favours recall, whereas damage and insurance assessment favour precision. Because the head outputs a probability, the operating point can be moved by calibrating the decision threshold, which is planned next.
- **Absolute scores are low compared with Sen1Floods11 benchmarks.** This is expected given the stricter event-wise protocol (every test flood is unseen), optical-only input, the coarse and partly generalised nature of NDEM polygons relative to pixel-level hand labels, and the small training set. The validation IoU (0.299) being lower than the test IoU (0.375) also reflects the small number of validation events (12) and inter-event variability.

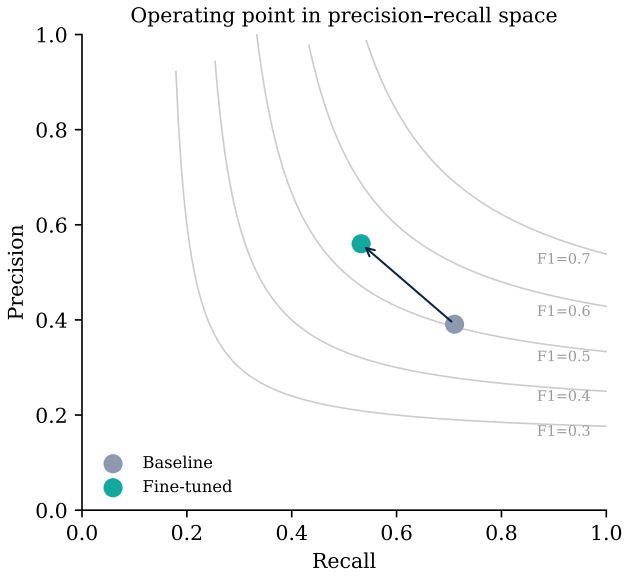


Figure 6.3: Baseline and fine-tuned operating points in precision–recall space with iso-F1 contours.

6.6 Pending analyses

The following analyses have been designed and are being prepared for the final report **[insert when available]**: per-event IoU, precision and recall for all 16 test events; confusion matrices; inference time per chip and per scene; and four-panel visual comparisons (Sentinel-2 false colour, NDEM truth, prediction, error map) for representative successes and failures.

CHAPTER 7

Physics-Based Flood Routing: Saint-Venant Equations and PINNs

7.1 Motivation

At the start of the project the team was asked to study the Saint-Venant equations and their applications, supported by a code base. Physics-based routing predicts how much water arrives where and when; satellite mapping observes where water actually went. Understanding both is necessary to integrate the observed flood extent into the ClimMaTech platform.

7.2 Governing equations

For one-dimensional unsteady flow in an open channel with lateral inflow q, the Saint-Venant equations in conservative form are

$$\frac{\partial A}{\partial t} + \frac{\partial Q}{\partial x} = q, \tag{7.1}$$

$$\frac{\partial Q}{\partial t} + \frac{\partial}{\partial x}\left(\frac{Q^2}{A}\right) + gA\frac{\partial h}{\partial x} = gA\,(S_0 - S_f), \tag{7.2}$$

with the friction slope given by Manning's equation,

$$S_f = \frac{n^2\,Q\,|Q|}{A^2 R^{4/3}}, \qquad R = \frac{A}{P}, \tag{7.3}$$

where P is the wetted perimeter. Equation (7.1) expresses conservation of mass and Equation (7.2) conservation of momentum (Newton's second law for a control volume), balancing local and convective acceleration, the pressure gradient, gravity and bed friction.

7.2.1 Approximations

Table 7.1: Hierarchy of flood-routing approximations of the momentum equation.

Model	Terms retained	Typical applicability
Kinematic wave	$S_0 = S_f$	Steep rivers; no backwater; no attenuation by pressure term
Diffusive wave	$S_0 - S_f - \partial h/\partial x = 0$	Captures attenuation; mild slopes
Dynamic wave	All terms	Flat reaches, tidal rivers, dam-break, backwater

7.2.2 Applications

The equations underpin flood forecasting and routing (predicting peak magnitude and arrival time downstream), dam-break and reservoir-release analysis, floodplain inundation modelling (e.g. HEC-RAS, LISFLOOD-FP), design of embankments and bridges, irrigation-canal operation, and the calibration of hydrological models against observations — including, in this project, satellite-observed flood extents.

7.3 Numerical solver developed in this project

To build intuition before using the PINN code, a compact explicit solver (`sve_demo.py`, Python/NumPy, Appendix A) was written for a wide rectangular prismatic channel. In vector form $\partial \mathbf{U}/\partial t + \partial \mathbf{F}/\partial x = \mathbf{S}$ with

$$\mathbf{U} = \begin{bmatrix} A \\ Q \end{bmatrix}, \quad \mathbf{F} = \begin{bmatrix} Q \\ Q^2/A + \frac{1}{2} g B h^2 \end{bmatrix}, \quad \mathbf{S} = \begin{bmatrix} 0 \\ gA(S_0 - S_f) \end{bmatrix}, \tag{7.4}$$

the MacCormack predictor–corrector scheme is

$$\mathbf{U}_i^* = \mathbf{U}_i^n - \frac{\Delta t}{\Delta x}\left(\mathbf{F}_{i+1}^n - \mathbf{F}_i^n\right) + \Delta t\, \mathbf{S}_i^n, \tag{7.5}$$

$$\mathbf{U}_i^{n+1} = \frac{1}{2}\left[\mathbf{U}_i^n + \mathbf{U}_i^* - \frac{\Delta t}{\Delta x}\left(\mathbf{F}_i^* - \mathbf{F}_{i-1}^*\right) + \Delta t\, \mathbf{S}_i^*\right], \tag{7.6}$$

which is second-order accurate in space and time. The time step is set adaptively by the CFL condition

$$\Delta t = C \frac{\Delta x}{\max_i \left(|u_i| + \sqrt{g h_i}\right)}, \qquad C = 0.8, \tag{7.7}$$

which gave $\Delta t \approx 24$–46 s for the conditions simulated.

Table 7.2: Parameters of the reference routing experiment (synthetic channel).

Parameter	Value	Parameter	Value
Reach length L	20 km	Grid spacing Δx	250 m (81 nodes)
Width B	50 m	Bed slope S_0	0.0005
Manning n	0.035	Base flow Q_b	50 $m^3\,s^{-1}$
Normal depth at Q_b	1.31 m	Peak inflow Q_p	400 $m^3\,s^{-1}$ at $t = 3$ h
Hydrograph	Gaussian, $\sigma = 40$ min	Simulated time	10 h
Upstream BC	Inflow hydrograph $Q(0,t)$	Downstream BC	Transmissive (zero gradient)
Initial condition	Uniform flow at Q_b	CFL number	0.8

7.3.1 Results

The simulated hydrographs (Figure 7.1) show the two classical features of flood-wave propagation: *attenuation* of the peak through channel storage and friction, and *translation* (delay) of the peak downstream. The peak fell from 400 to 314 $m^3\,s^{-1}$ at 8 km (arriving 0.76 h later) and to 230 $m^3\,s^{-1}$ at 20 km (2.25 h later), i.e. a mean wave celerity of about 2.5 $m\,s^{-1}$. Part of the attenuation is attributable to numerical diffusion of the scheme, and the outlet node is affected by the downstream boundary condition; the experiment is therefore illustrative rather than a validated simulation.

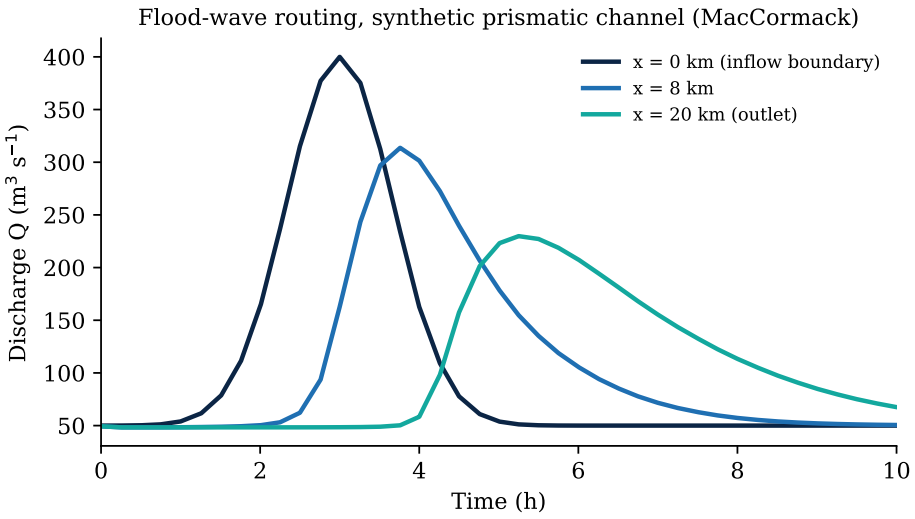


Figure 7.1: Simulated discharge hydrographs at the inflow, 8 km and 20 km.

7.3.2 Parameter sensitivity

Single-parameter perturbations were run to understand how uncertain channel properties — the very quantities that are costly to survey for the platform — affect flood predictions (Table 7.3, Figure 7.2).

Table 7.3: Sensitivity of peak discharge and travel time at 20 km to channel and flood parameters.

Scenario	Peak at 8 km ($m^3\ s^{-1}$)	Peak at 20 km ($m^3\ s^{-1}$)	Attenuation at 20 km (%)	Travel time to 20 km (h)
Reference	313.7	229.8	42.6	2.25
Rougher bed, $n = 0.050$	276.7	186.1	53.5	3.00
Steeper bed, $S_0 = 0.001$	355.8	303.4	24.2	2.00
Wider channel, $B = 100$ m	298.0	215.2	46.2	3.26
Larger flood, $Q_p = 800\ m^3\ s^{-1}$	643.2	460.0	42.5	1.75

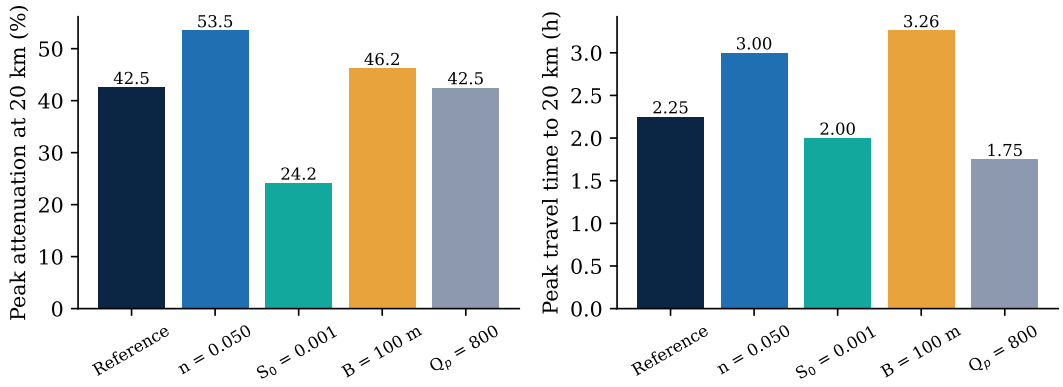


Figure 7.2: Peak attenuation and travel time to 20 km for each scenario.

Three observations follow. First, roughness and slope — both uncertain in practice — change the downstream peak by -19% and $+32\%$ respectively, so errors in these inputs propagate directly into flood forecasts. Second, a larger flood travels faster (1.75 h versus 2.25 h), so warning time shrinks precisely when it matters most. Third, wider channels store more water and delay the peak. These results quantify why an independent observation of flood extent is valuable for constraining physically based models.

7.4 Physics-informed neural network code base

The mentors provided a PINN code base for the Saint-Venant equations (`https://github.com/Barbhuiya12/PINN`). A PINN approximates $(A, Q) = \mathcal{N}_\theta(x,t)$ with a fully connected network and trains θ to minimise

$$\mathscr{L}(\theta) = \underbrace{\frac{1}{N_d}\sum \left\| Q_\theta - Q^{\text{obs}} \right\|^2}_{\text{data}} + \lambda_r \underbrace{\frac{1}{N_r}\sum \left(r_{\text{mass}}^2 + r_{\text{mom}}^2 \right)}_{\text{PDE residual}} + \lambda_b \underbrace{\frac{1}{N_b}\sum \left\| \mathbf{U}_\theta - \mathbf{U}^{\text{BC/IC}} \right\|^2}_{\text{boundary/initial}}, \quad (7.8)$$

where r_{mass} and r_{mom} are the left-minus-right-hand sides of Equations (7.1) and (7.2), computed by automatic differentiation at collocation points (Figure 7.3).

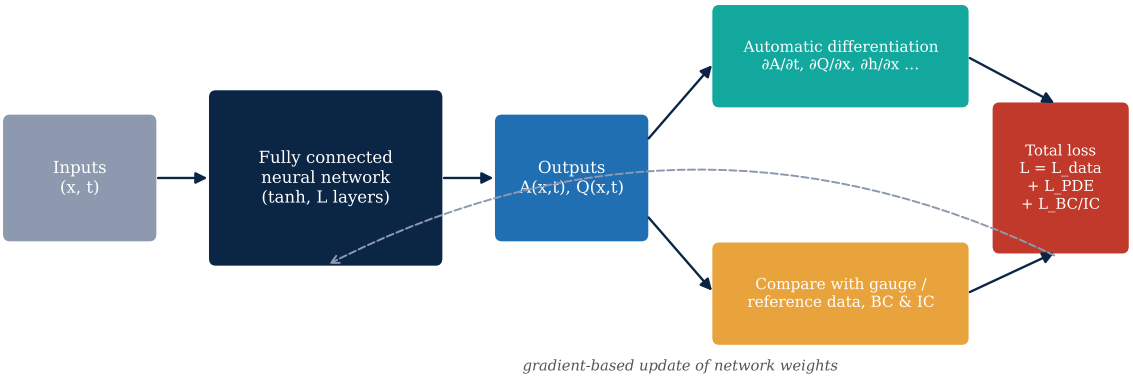


Figure 7.3: Structure of a physics-informed neural network for the Saint-Venant equations.

Table 7.4: Workflow of the mentor-provided PINN code base.

Stage	Description
Setup	`bash SETUP.sh` creates the isolated conda environment `pinn_sve` (Python 3.7, TensorFlow 1.15; not TF2-compatible)
Verification	`python phase4_validation/validate.py -case 3` must report a deviation of 0.0%
Phase 1: data exploration	`p1a_read_reference.py`, `p1b_build_training_sets.py`, `p1c_explore_plots.py` with `-case N`
Training	PINN training on the assigned river reach
Phase 4: validation	Comparison of the PINN solution with the reference solution
Cases	Six cases, each a different river reach (`docs/02_case_assignments.md`)

Status: **[state completed stages and assigned case numbers, e.g. setup verified; Phase 1 completed for case N]**.

CHAPTER 8

Inference Workflow, Applications and Field Visit

8.1 Inference workflow

The deployed model is intended to process a new Sentinel-2 scene acquired during an event and return a georeferenced flood-extent product (Figure 8.1):

1. Acquire the Sentinel-2 L2A scene over the area of interest.
2. Select and order the six bands; apply the scaling and normalisation used in training.
3. Mask cloud, cloud shadow and permanent water.
4. Tile the scene into 512×512 chips and run Prithvi Flood-IN on each chip.
5. Mosaic the per-pixel flood probabilities, apply the calibrated decision threshold, and write a georeferenced raster (GeoTIFF) and vector polygons.
6. Overlay the flood extent on dams and reservoir releases, river-gauge readings, roads and bridges, villages and critical infrastructure (hospitals, shelters, substations) in a decision-support dashboard.

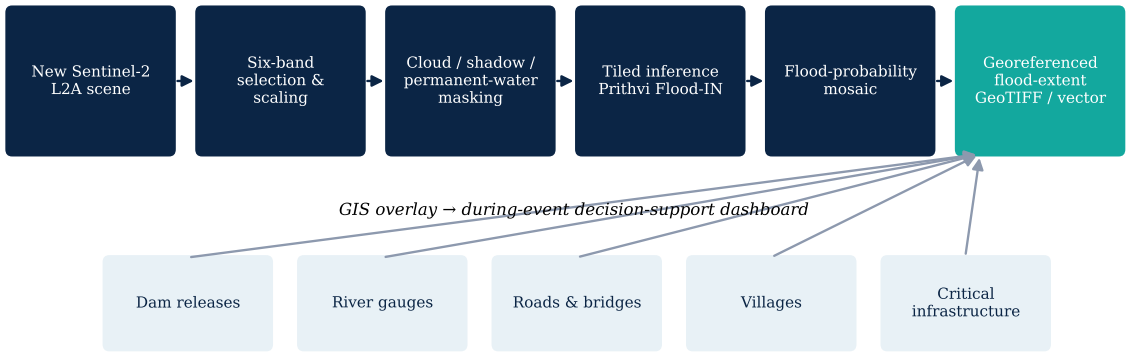


Figure 8.1: Inference and decision-support workflow.

8.2 Applications

Table 8.1: Application pathways for Prithvi Flood-IN.

Application	Description
Mahanadi basin HIS	Near-real-time inundation layer combined with Hirakud reservoir releases and CWC gauge data within the ClimMaTech–IITB platform
Disaster response	Flood-extent maps for NDMA/SDMAs and district control rooms: evacuation planning, relief routing, identification of cut-off villages and roads
Agriculture and insurance	Estimation of inundated cropland per district to accelerate crop-loss assessment and claims under schemes such as PMFBY
Infrastructure risk	Flagging of flooded roads, bridges, substations, hospitals and schools; prioritisation of restoration
Hydrological model validation	Observed extents to calibrate and validate hydrodynamic and rainfall–runoff simulations (Chapter 7)
Flood-risk planning	Multi-year inundation-frequency maps for zoning, embankment planning and climate-risk studies

The precision–recall balance of the model should be chosen per application: higher recall is preferred for early warning and evacuation, where a missed flood is costlier than a false alarm, whereas higher precision is preferred for damage and insurance assessment.

8.3 Field visit: Pawna Dam



Figure 8.2: Project team at Pawna Dam, Maval, Pune district, Maharashtra, during the monsoon **[date of visit]**.

The team visited Pawna Dam on the Pawna river in Maval taluka, Pune district, during the monsoon **[date]**. The purpose was to observe a monsoon-fed reservoir in operation: the dam structure and spillway, the monitoring of reservoir water levels, and how release decisions are taken **[add specific observations from the visit, e.g. gate operation, level gauges, release rules]**. Three lessons from the visit inform the project design:

1. Reservoir release decisions directly shape where and when flooding occurs downstream, so dam data must be a core overlay layer in the dashboard; for the Mahanadi pilot this corresponds to Hirakud releases.
2. Satellite flood extents are most informative when combined with reservoir levels and gauge readings.
3. The heavy monsoon cloud over the reservoir illustrated directly why optical imagery alone is insufficient during floods, motivating Sentinel-1 SAR fusion.

8.4 Industry visit (proposed)

A visit to ClimMaTech Pvt. Ltd. (IIT Mandi, Kamand Campus, District Mandi, Himachal Pradesh – 175075) is proposed **[dates to be confirmed with mentor]** to review the hydrological platform with the product team, demonstrate Prithvi Flood-IN, and agree on the

integration of flood maps and the Mahanadi pilot.

CHAPTER 9

Discussion and Limitations

9.1 Key findings so far

1. Fine-tuning a global flood foundation model on 100 Indian events measurably improves it on unseen Indian floods (IoU +11%, precision +43%), primarily by suppressing false alarms.
2. The improvement comes with reduced recall; the model's operating point must be chosen for the application, and threshold calibration is the cheapest lever.
3. In the controlled benchmark, a multimodal foundation model outperformed a from-scratch U-Net by about 0.04 IoU, but the U-Net was strong and label-efficient, so a U-Net baseline must be included for the Indian dataset before drawing firm conclusions.
4. Channel roughness and slope, the uncertain inputs of the existing platform, strongly influence routed peaks, which supports the value of observed flood extents for model constraint.

9.2 Limitations

Table 9.1: Limitations of the current work and planned mitigations.

Limitation	Effect	Planned mitigation
Small dataset (100 events)	Limited diversity of regions, seasons and land cover	Screen about 2,680 further NDEM layers
Optical-only input	Cloudy monsoon scenes unusable; many NDEM maps discarded	Sentinel-1 SAR fusion (e.g. TerraMind-style multimodal input)
Reduced recall (0.71 $\rightarrow$ 0.53)	More missed flood area	Threshold calibration; class-weighted or Dice loss
Label quality	NDEM polygons are generalised; boundary errors penalise IoU	Boundary-tolerant metrics; visual audit
No from-scratch baseline on Indian data	Value of pre-training on Indian data not yet isolated	Train U-Net on the same split
Single training run	No estimate of run-to-run variance	Repeat runs with different seeds
Free-tier compute	Disconnects, memory limits, slow iteration	Dedicated GPU via mentors
Synthetic SVE experiment	Not validated on a real reach	PINN cases and gauge data

CHAPTER 10

Work Plan for the Second Half

10.1 Milestones

Table 10.1: Project milestones and status as of 5 October 2026.

No.	Milestone	Target	Status
M1	Literature review; platform and Prithvi-EO-2.0 study	Aug 2026	Completed
M2	Saint-Venant study; PINN setup and Phase 1	Aug–Sep 2026	**[confirm]**
M3	Indian flood dataset (100 events, 3,584 chips)	Sep 2026	Completed
M4	Baseline evaluation and full fine-tuning	Sep 2026	Completed
M5	Benchmark study (Sen1Floods11)	Sep 2026	Completed
M6	Error analysis, threshold calibration, U-Net baseline, FastFlood	Oct 2026	Planned
M7	Dataset expansion and Sentinel-1 SAR fusion; PINN training	Oct–Nov 2026	Planned
M8	Mahanadi pilot and dashboard prototype	Nov 2026	Planned
M9	Final report, code, model and presentation	**[end-semester date]**	Planned

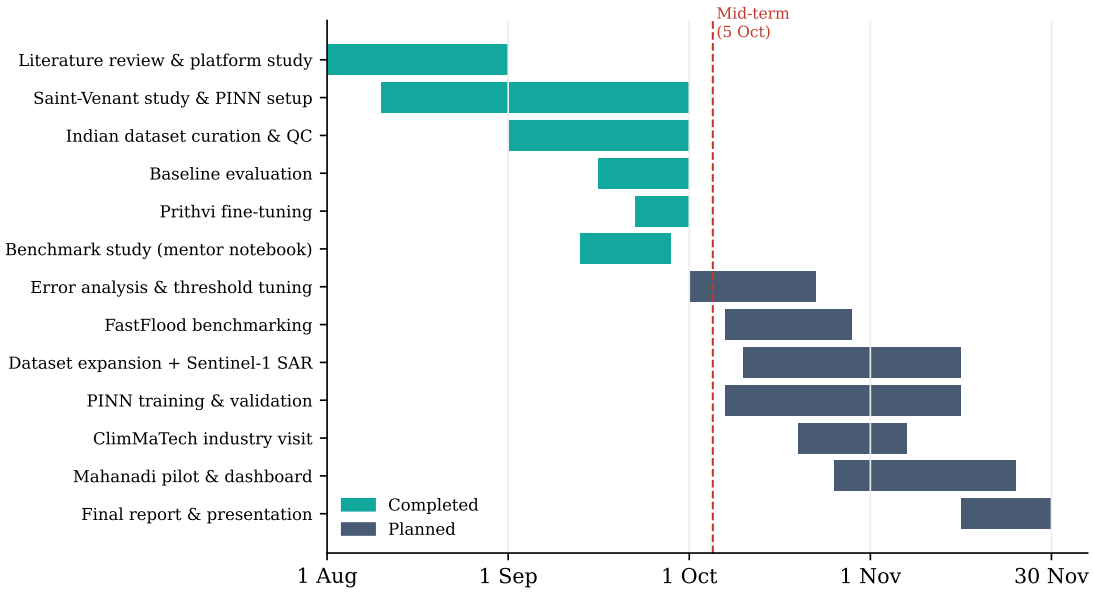


Figure 10.1: Project timeline (Gantt chart).

10.2 Planned technical work

1. **Error analysis:** per-event metrics, confusion matrices and error maps to identify failure modes (turbid water, urban areas, cloud edges, flooded vegetation).
2. **Threshold calibration:** sweep the flood-probability threshold on the validation set and report precision–recall and F1 curves; choose application-specific operating points.
3. **Loss design:** evaluate class-weighted cross-entropy and Dice loss to recover recall.
4. **Baselines:** train a U-Net from scratch on the Indian split under the same protocol, so that the benefit of pre-training is isolated.
5. **Data expansion and SAR fusion:** screen the remaining NDEM layers; add Sentinel-1 VV/VH to recover cloudy events.
6. **External benchmarking:** compare against FastFlood validation extents where available.
7. **PINN track:** complete PINN training and validation on the assigned river reaches and compare with the finite-difference solver.
8. **Mahanadi pilot:** apply the model to recent Mahanadi events and overlay with Hirakud releases and CWC gauges in a dashboard prototype.

10.3 Expected deliverables

- India-specific flood segmentation dataset with documented curation procedure.
- Fine-tuned Prithvi Flood-IN checkpoint, training and evaluation code.
- Evaluation report with per-event metrics and error maps.
- Inference pipeline producing georeferenced flood-extent maps.
- Dashboard prototype for the Mahanadi basin.
- Saint-Venant and PINN study with results on assigned reaches.
- Final report and presentation.

10.4 Roles and responsibilities

All members share common work: understanding Prithvi-EO-2.0 and its training methodology, participating in reviews, and contributing to reports and presentations.

Table 10.2: Primary responsibilities of team members.

Member	Primary responsibilities
Nikhil Upadhyay (Leader)	Coordination with supervisor, TA and ClimMaTech; research on applying the fine-tuned model for flood prediction and analysis; dashboard design and Mahanadi pilot; reports and presentations
Prakhar Gupta	Data used to train Prithvi; Indian fine-tuning dataset: scene–label pairing, masking, event-wise split, storage optimisation
Prabhat Kumar	Applications of the fine-tuned model; Saint-Venant study, routing experiments and PINN code base; presentations
Vineet Jangir	Training methodology of foundation and fine-tuned models; execution of model training, environment and checkpoint management
Himanshu Shete	Performance evaluation (IoU, precision, recall, F1, confusion matrix, inference time); baseline comparison and error maps; inference workflow

10.5 Risks

Table 10.3: Project risks and mitigation.

Risk	Likelihood	Mitigation
Compute interruptions on free tier	High	Local-first checkpointing; request dedicated GPU from mentors
Few additional cloud-free scenes	Medium	Sentinel-1 SAR fusion
NDEM access or format changes	Low	Cache downloaded layers; document provenance
End-semester workload	Medium	Front-load experiments in October

CHAPTER 11

Conclusions

In the first half of the project the team has built an India-specific flood segmentation dataset from official NDEM maps and Sentinel-2 imagery under a strict event-wise protocol, fully fine-tuned the 300 M-parameter Prithvi-EO-2.0 flood model on free-tier infrastructure, and shown on 16 unseen flood events that fine-tuning improves IoU by 11% and precision by 43%, at the cost of lower recall. A mentor-provided benchmark confirmed that a multimodal foundation model can outperform strong from-scratch baselines, while also showing how strong and label-efficient a U-Net can be. On the physics side, the team implemented and analysed a Saint-Venant routing solver, quantified the sensitivity of flood peaks to uncertain channel parameters, and set up a PINN framework. Together with the inference workflow, the application analysis and the Pawna Dam field visit, this establishes the foundation for the second half: calibrating the model's operating point, adding radar data, validating against independent benchmarks and piloting an integrated flood information product for the Mahanadi basin.

References

[1] D. Bonafilia, B. Tellman, T. Anderson and E. Issenberg, "Sen1Floods11: A georeferenced dataset to train and test deep learning flood algorithms for Sentinel-1," in *Proc. IEEE/CVF CVPR Workshops*, 2020, pp. 835–845.

[2] D. Szwarcman *et al.*, "Prithvi-EO-2.0: A versatile multi-temporal foundation model for Earth observation applications," arXiv:2412.02732, 2024.

[3] J. Jakubik *et al.*, "Foundation models for generalist geospatial artificial intelligence," arXiv:2310.18660, 2023.

[4] J. Jakubik *et al.*, "TerraMind: Large-scale generative multimodality for Earth observation," arXiv:2504.11171, 2025.

[5] C. Gomes *et al.*, "TerraTorch: The geospatial foundation models toolkit," arXiv:2503.20563, 2025.

[6] V. Marsocci *et al.*, "PANGAEA: A global and inclusive benchmark for geospatial foundation models," arXiv:2412.04204, 2024.

[7] Kaushik *et al.*, study on hyper-parameter sensitivity of geospatial foundation-model fine-tuning, 2025, as cited in [15]. **[complete reference]**

[8] N. Otsu, "A threshold selection method from gray-level histograms," *IEEE Trans. Systems, Man, and Cybernetics*, vol. 9, no. 1, pp. 62–66, 1979.

[9] O. Ronneberger, P. Fischer and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *MICCAI*, LNCS 9351, 2015, pp. 234–241.

[10] A. Dosovitskiy *et al.*, "An image is worth 16×16 words: Transformers for image recognition at scale," in *ICLR*, 2021.

[11] T. Xiao, Y. Liu, B. Zhou, Y. Jiang and J. Sun, "Unified perceptual parsing for scene understanding," in *ECCV*, 2018.

[12] V. T. Chow, D. R. Maidment and L. W. Mays, *Applied Hydrology*. New York: McGraw-Hill, 1988.

[13] M. Raissi, P. Perdikaris and G. E. Karniadakis, "Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations," *J. Computational Physics*, vol. 378, pp. 686–707, 2019.

[14] A. Paszke *et al.*, "PyTorch: An imperative style, high-performance deep learning library," in *NeurIPS*, vol. 32, 2019, pp. 8024–8035.

[15] S. Barbhuiya and V. Gupta, "Does a foundation model beat a U-Net at flood mapping?," Jupyter notebook, IIT Mandi / ClimMaTech, 2026. Available: `https://github.com/Barbhuiya12/Flood_Resource`

[16] S. Barbhuiya, "PINN for the Saint-Venant equations," code repository, 2026. Available: `https://github.com/Barbhuiya12/PINN`

[17] National Remote Sensing Centre (ISRO), *National Database for Emergency Management (NDEM)*, flood inundation layers, 2021–2026.

[18] FastFlood, "Global validation," `https://fastflood.org/global/build/validation`, accessed Sept. 2026.

[19] European Space Agency, *Copernicus Sentinel-2 Level-2A surface reflectance products*, 2021–2026.

CHAPTER A

Code Listings

A.1 Saint-Venant MacCormack solver (`sve_demo.py`)

```
"""1-D Saint-Venant flood routing demo (MacCormack scheme), wide rectangular channel
    .
Illustrative only: synthetic channel, not calibrated to a real river."""
import numpy as np, json
g=9.81; L=20000; dx=250; B=50.0; S0=5e-4; n=0.035
x=np.arange(0,L+dx,dx); N=len(x)
def normal_depth(Q):
    h=1.0
    for _ in range(100):  # Manning, wide channel: Q = B h^(5/3) sqrt(S0)/n
        h=(Q*n/(B*np.sqrt(S0)))**0.6
    return h
Qb,Qp,tp,sig=50.0,400.0,3*3600,2400.0
inflow=lambda t: Qb+(Qp-Qb)*np.exp(-0.5*((t-tp)/sig)**2)
h0=normal_depth(Qb); A=np.full(N,B*h0); Q=np.full(N,Qb)
def flux(A,Q):
    h=A/B; return Q, Q**2/A+g*B*h**2/2
def src(A,Q):
    h=A/B; R=A/(B+2*h); Sf=n**2*Q*np.abs(Q)/(A**2*R**(4/3))
    return np.zeros_like(A), g*A*(S0-Sf)
t=0; T=10*3600; stations={0:[],8000:[],20000:[]}; times=[]
idx={k:int(k/dx) for k in stations}; nextrec=0
while t<T:
    c=np.max(np.abs(Q/A)+np.sqrt(g*A/B)); dt=0.8*dx/c
    F1,F2=flux(A,Q); S1,S2=src(A,Q)
    Ap=A.copy(); Qp_=Q.copy()
    Ap[:-1]=A[:-1]-dt/dx*(F1[1:]-F1[:-1])+dt*S1[:-1]
    Qp_[:-1]=Q[:-1]-dt/dx*(F2[1:]-F2[:-1])+dt*S2[:-1]
    G1,G2=flux(Ap,Qp_); T1,T2=src(Ap,Qp_)
    An=A.copy(); Qn=Q.copy()
    An[1:]=0.5*(A[1:]+Ap[1:]-dt/dx*(G1[1:]-G1[:-1])+dt*T1[1:])
    Qn[1:]=0.5*(Q[1:]+Qp_[1:]-dt/dx*(G2[1:]-G2[:-1])+dt*T2[1:])
    t+=dt; Qn[0]=inflow(t); An[0]=An[1]         # upstream: inflow hydrograph
    An[-1]=An[-2]; Qn[-1]=Qn[-2]                 # downstream: transmissive
    A,Q=An,Qn
    if t>=nextrec:
        times.append(round(t/3600,2)); [stations[k].append(float(Q[idx[k]])) for k
    in stations]; nextrec+=900
out={'t_h':times,'Q':{str(k):v for k,v in stations.items()}}
```

```
for k,v in stations.items():
    i=int(np.argmax(v)); print(f"x={k/1000:>4.0f} km  peak Q={max(v):6.1f} m3/s at t
    ={times[i]:.2f} h")
json.dump(out,open('sve_demo.json','w'))
```

A.2 Masked loss for unlabelled pixels (from the benchmark notebook)

```
counts_in_loss = (labels >= 0).float()      # 1 water, 0 dry, -1 unlabelled
targets = labels.clamp(min=0)
predictions = model(images).squeeze(1)
loss_per_pixel = nn.functional.binary_cross_entropy_with_logits(
    predictions, targets, reduction="none")
loss = (loss_per_pixel * counts_in_loss).sum() / counts_in_loss.sum()
```

A.3 Saint-Venant PDE residual for a PINN (illustrative sketch, TensorFlow 1.x)

```
def sve_residual(net, x, t):
    A, Q = net(tf.concat([x, t], 1))
    h   = A / B
    A_t = tf.gradients(A, t)[0]
    Q_t = tf.gradients(Q, t)[0]
    Q_x = tf.gradients(Q, x)[0]
    F_x = tf.gradients(Q**2 / A, x)[0]
    h_x = tf.gradients(h, x)[0]
    R   = A / (B + 2*h)
    Sf  = n**2 * Q * tf.abs(Q) / (A**2 * R**(4/3))
    r_mass = A_t + Q_x                                  # continuity
    r_mom  = Q_t + F_x + g*A*h_x - g*A*(S0 - Sf)        # momentum
    return r_mass, r_mom
# loss = mse(Q_pred, Q_obs) + mse(r_mass, 0) + mse(r_mom, 0) + mse(bc_pred, bc_obs)
```

CHAPTER B

Project Log

Date (2026)	Activity
29 Jul	Team formed; registration for the IDP
3 Aug	Kick-off with Prof. Biswal and TA; walkthrough of the existing platform and its data gaps
5 Aug	Meeting at supervisor's office; reference material shared
6–7 Aug	ClimMaTech team joins the project group; review meeting (Room 218)
11 Aug	Progress meeting (online)
12 Aug	PINN–Saint-Venant code base and Phase-1 instructions shared by mentors
14 Aug	Review meeting; reading material on inundation, routing and runoff generation shared
21 Aug	Progress meeting
25–26 Aug	Objectives finalised; DUGC proposal form submitted
28 Aug	Mentor directs team to TerraTorch, Prithvi-EO-2.0 and TerraMind examples
6 Sep	Indian flood fine-tuning catalogue shared with mentors for review
10 Sep	FastFlood validation benchmark shared by TA
15 Sep	Mentor benchmark notebook (TerraMind vs U-Net) executed
[date]	Field visit to Pawna Dam
23 Sep	Fine-tuning complete; progress review with supervisor and mentors
5 Oct	Mid-term report

CHAPTER C

Data and Code Locations

Item	Location
Training and evaluation code	`scripts/finetune.py`, `scripts/evaluate.py`, `floodin/model.py`, `floodin/data.py`
Dataset selection	`data/selection/selection.csv`
Training data	Google Drive: `prithvi-floodin/data/`
Pre-trained checkpoint	Google Drive: `prithvi-floodin/checkpoints/`
Fine-tuned checkpoint	Google Drive: `prithvi-floodin/run1/checkpoints/best-07.ckpt`
Result files	Google Drive: `prithvi-floodin/results/baseline_test.csv`, `finetuned_test.csv`
Saint-Venant solver	`sve_demo.py`
Benchmark notebook	`https://github.com/Barbhuiya12/Flood_Resource`
PINN code base	`https://github.com/Barbhuiya12/PINN`